

	2. Copies / number of mRNA related to amount of enzyme / amoA produced / translated; 3. Don't know how much mRNA / amoA produced by each cell; 4. Don't know if amoA (mRNA / enzyme) is linked to cell division / growth (of population);		factors / other named factors affecting growth 3. Accept some bacteria produce more mRNA / amoA than others 4. Amount of amoA does not show cell division / growth 4. Reject references to mitosis / meiosis
06.6	1. Suitable method; eg in boiling water / steam / autoclave / wash in disinfectant / wash in alcohol 2. (Reason) to remove / kill other bacteria / organisms that might break down ammonia;	2	1. Ignore heat unqualified 1. Ignore flaming of bottle 1. Accept radiation 2. Ignore ref to removing bacteria that 'affect the result' 2. Accept other bacteria producing amoA 2. Accept other bacteria compete with / kill bacteria that produce amoA 2. Ignore contamination unqualified

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

21– 25	Extended Abstract Generalised beyond specific context	Response shows holistic approach to the question with a fully integrated answer which makes clear links between several different topics and the theme of the question. Biology is detailed and comprehensive A-level content, uses appropriate terminology, and is very well written and always clearly explained. No significant errors or irrelevant material. For top marks in the band, the answer shows evidence of reading beyond specification requirements.
16– 20	Relational Integrated into a whole	Response links several topics to the main theme of the question, to form a series of interrelated points which are clearly explained. Biology is fundamentally correct A-level content and contains some points which are detailed, though there may be some which are less well developed, with appropriate use of terminology. Perhaps one significant error and, or, one irrelevant topic which detracts from the overall quality of the answer.
11– 15	Multistructural Several aspects covered but they are unrelated	Response mostly deals with suitable topics but they are not interrelated and links are not made to the theme of the question. Biology is usually correct A-level content, though it lacks detail. It is usually clearly explained and generally uses appropriate terminology. Some significant errors and, or, more than one irrelevant topic.
6– 10	Unistructural Only one or few aspects covered	Response predominantly deals with only one or two topics that relate to the question. Biology presented shows some superficial A-level content that may be poorly explained, lacking in detail, or show limited use of appropriate terminology. May contain a number of significant errors and, or, irrelevant topics.
1–5	Unfocused	Response only indirectly addresses the theme of the question and merely presents a series of biological facts which are usually descriptive in nature or poorly explained and at times may be factually incorrect. Content and terminology is generally below A-level. May contain a large number of errors and, or, irrelevant topics.
0		Nothing of relevance or no response.

Commentary on terms and statements in the levels mark scheme

The levels mark scheme for the essay contains a number of words and statements that are open to different interpretations. This commentary defines the meanings of these words and statements in the context of marking the essay. Many words and statements are used in the descriptions of more than one level of response. The definitions of these remain the same throughout.

Levels mark scheme word/statement	Definition
Holistic	Synoptic, drawing from different topics (usually sections of the specification)
A fully integrated answer which makes clear links between several different topics and the theme of the question	All topics relate to the title and theme of the essay; for example, explaining the biological importance of a process. When considering, for example, the importance of a process, the explanation must be at A-level standard. ‘Several’ here is defined as at least four topic areas from the specification covered. This means some sentences, not just a word or two. It does not mean using many examples from one topic area.
Biology is detailed and comprehensive A-level content, uses appropriate terminology, and is very well written and always clearly explained.	Detailed and comprehensive A-level content is the specification content. Terminology is that used in the specification. Well written and clearly explained refers mainly to biological content and use of terminology. Prose, handwriting and spelling are secondary considerations. Phonetic spelling is accepted, unless examiners are instructed not to do so for particular words; for example, glucagon, glucose and glycogen.
No significant errors or irrelevant material.	A significant error is one which significantly detracts from the biological accuracy or correctness of a described example. This will usually involve more than one word. Irrelevant material is several lines (or more) that clearly fails to address the title, or the theme of the title.
For top marks in the band, the answer shows evidence of reading beyond specification requirements.	An example that is relevant to the title and is not required in the specification content. The example must be used at A-level standard.
Response mostly deals with suitable topics but they are not interrelated and links are not made to the theme of the question.	Not addressing the biological theme of the essay (eg importance) <u>at A-level standard</u> .

Question	Marking Guidance	Mark	
07.1	The importance of the control of movement in cells and organisms. Suitable topic areas • 3.1.4.2 Enzymes and control of action • 3.1.5.2 DNA replication • 3.2.2 Mitosis, binary fission • 3.2.3 Transport across membranes • 3.2.4 Cell recognition and the immune system • 3.3.2 Gas exchange • 3.3.3 Digestion and absorption • 3.3.4.1 Mass transport in animals • 3.3.4.2 Mass transport in plants • 3.4.2 DNA and protein synthesis • 3.4.3 Meiosis • 3.5.1 Photosynthesis • 3.5.2 Respiration • 3.6.1.1 Survival and response • 3.6.1.2 Receptors • 3.6.1.3 Control of heart rate • 3.6.2.1 Nervous impulses • 3.6.2.2 Synaptic transmission • 3.6.3 Muscle contraction • 3.6.4.2 Control of blood glucose • 3.6.4.3 Control of blood water potential • 3.7.1 Inheritance • 3.8.2.2 Regulation of transcription and translation • 3.8.2.3 Gene expression and cancer	[25 marks]	

In order to fully address the question and reach the highest mark bands students must also include at least four topics in their answer, to demonstrate a synoptic approach to the essay.

Students may be able to show the relevance of other topics from the specification.

Note, other topics from beyond the specification can be used, providing they relate to the title and contain factually correct material of at least an A-level standard. Credit should not be given for topics beyond the specification which are below A-level standard.

Question	Marking Guidance	Mark
07.2	The importance of interactions between cells and between organisms. Suitable topic areas • 3.2.1.2 Viruses • 3.2.4 Cell recognition, immune system, HIV • 3.3.2 Gas exchange • 3.3.4.1 Mass transport in animals • 3.3.4.2 Mass transport in plants • 3.4.4 Genetic diversity and adaptation • 3.4.5 Species and taxonomy (courtship behaviour) • 3.4.6 Biodiversity within a community • 3.5.3 Energy and ecosystems • 3.5.4 Nutrient cycles • 3.6.1.1 Survival and response • 3.6.1.2 Receptors • 3.6.2.2 Synaptic transmission • 3.6.3 Skeletal muscles are stimulated ... • 3.6.4.2 Control of blood glucose • 3.6.4.3 Control of blood water potential • 3.7.1 Inheritance • 3.7.2 Populations in ecosystems • 3.7.3 Evolution and speciation • 3.7.4 Populations in ecosystems • 3.8.2.3 Gene expression and cancer	[25 marks]

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